https://www.programiz.com/online-compiler/3Z2tG4tDYcnhC

Description

Create a Student Management System using Arrays in C#. The program should allow storing student names and grades, displaying them, calculating averages, and practicing different types of arrays.

Part 1 – Student Names (One Dimensional Array)

Create an array that stores 5 student names. Ask the user to enter the names, store them in the array, and print them using a loop.

```
using System;

class StudentManagement
{
    static void Main()
    {
        // ======================
        // Part 1: Student Names
        // ======================
        string[] studentNames = new string[5];

        Console.WriteLine("Enter names of 5 students:");
        for (int i = 0; i < 5; i++)
        {
            Console.Write("Student " + (i + 1) + ": ");
            studentNames[i] = Console.ReadLine();
        }

        Console.WriteLine("\nStudent Names:");
        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine(studentNames[i]);
        }
```

```
Enter names of 5 students:
Student 1: e
Student 2: b
Student 3: c
Student 4: a
Student 5: d

Student Names:
e
b
c
a
d

=== Code Execution Successful ===
```

Part 2 – Student Grades (Multi-Dimensional Array)

Create a 2D array for grades: 5 students and 3 subjects (Math, Science, English). Ask the user to enter the grades for each student and store them.

Main.cs Share Run

```
        // }

        // ======================
        // Part 2: 2D Grades
        // ======================
        int[,] grades = new int[5, 3];
        string[] subjects = { "Math", "Science", "English" };

        Console.WriteLine("\nEnter grades:");

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine("\nGrades for " + studentNames[i]);

            for (int j = 0; j < 3; j++)
            {
                Console.Write(subjects[j] + ": ");
                grades[i, j] = int.Parse(Console.ReadLine());
            }
        }
```

Output Clear

```
Enter grades:

Grades for
Math: 90
Science: 80
English: 90

Grades for
Math: 80
Science: 80
English: 80

Grades for
Math: 70
Science: 80
English: 0

Grades for
Math: 80
Science: 70
English: 90

Grades for
Math: 89
Science: 99
English: 88

=== Code Execution Successful ===
```

Part 3 – Display Student Grades

Print the grades in a table format showing each student and their grades.

Part 4 – Calculate Student Average

Calculate and display the average grade for each student.

Main.cs

```
        // ======================
        // Part 2: 2D Grades
        // ======================
        int[,] grades = new int[5, 3];
        string[] subjects = { "Math", "Science", "English" };

        Console.WriteLine("\nEnter grades:");

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine("\nGrades for " + studentNames[i]);

            for (int j = 0; j < 3; j++)
            {
                Console.Write(subjects[j] + ": ");
                grades[i, j] = int.Parse(Console.ReadLine());
            }
        }

//          // ======================
//          // Part 3: Display Grades
//          // ======================
        Console.WriteLine("\nStudent Grades:");
        Console.WriteLine("Name\tMath\tScience\tEnglish");

        for (int i = 0; i < 5; i++)
        {
            Console.Write(studentNames[i] + "\t");

            for (int j = 0; j < 3; j++)
            {
                Console.Write(grades[i, j] + "\t");
            }

            Console.WriteLine();
        }

//          // ======================
```

Output

```
Enter grades:

Grades for
Math: 90
Science:
English: 80

Grades for
Math: 88
Science: 99
English: 8

Grades for
Math: 77
Science: 8
English: 9

Grades for
Math: 88
Science: 88
English: 9

Grades for
Math: 99
Science: 99
English: 99

Student Grades:
Name    Math    Science English
    90  0   80
    88  99  88
    77  88  99
    88  88  99
    99  99  99

=== Code Execution Successful ===
```

```
//          // ======================
//          // Part 4: Averages
//          // ======================
        Console.WriteLine("\nStudent Averages:");

        for (int i = 0; i < 5; i++)
        {
            int sum = 0;

            for (int j = 0; j < 3; j++)
            {
                sum += grades[i, j];
            }

            double avg = (double)sum / 3;
            Console.WriteLine(studentNames[i] + ": " + avg);
        }
```

```
Student Averages:
oman: 80
ebtesam: 85.6666666666667
amani: 88
maryam: 80
noor: 85.6666666666667
```

Part 5 – Jagged Array

Create a jagged array where each student can have a different number of subjects. Display the number of subjects and grades for each student.

```
Enter jagged array data:

How many subjects for ww? 3
Grade 1: 100
Grade 2: 90
Grade 3: 80

How many subjects for ee? 3
Grade 1: 100
Grade 2: 80
Grade 3: 90

How many subjects for aa? 3
Grade 1: 80
Grade 2: 90
Grade 3: 90

How many subjects for bb? 3
Grade 1: 80
Grade 2: 0
Grade 3: 90

How many subjects for cc? 3
Grade 1: 80
Grade 2: 0
Grade 3: 90

Jagged Array Data:
ww has 3 subjects
100 90 80
ee has 3 subjects
100 80 90
aa has 3 subjects
80 90 90
bb has 3 subjects
80 70 90
cc has 3 subjects
80 0 90
```

Part 6 – Search for Student

Ask the user to enter a student name. If found, print the index of the student. If not found, display a message saying the student was not found.

```
//          // =====================
//          // Part 6: Search Student
//          // =====================
        Console.Write("\nEnter student name to search: ");
        string search = Console.ReadLine();

        bool found = false;

        for (int i = 0; i < 5; i++)
        {
            if (studentNames[i].Equals(search, StringComparison.OrdinalIgnoreCase))
            {
                Console.WriteLine("Student found at index: " + i);
                found = true;
                break;
            }
        }

        if (!found)
        {
            Console.WriteLine("Student not found.");
        }
```

```
Enter student name to search: E
Student found at index: 4
```

Part 7 – Find Highest Grade

Search through all grades and display the highest grade in the system.

```
// //          // ======================
// //          // Part 7: Highest Grade
// //          // ======================
        int highest = grades[0, 0];

        for (int i = 0; i < 5; i++)
        {
            for (int j = 0; j < 3; j++)
            {
                if (grades[i, j] > highest)
                {
                    highest = grades[i, j];
                }
            }
        }

        Console.WriteLine("\nHighest grade: " + highest);


    }
}
```

```
Highest grade: 100
```